

This assignment will not be graded and is only for practice.

Multiple Choice Questions (MCQ):

1) Which of the following statements are correct?

1 point

☐ There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(2, 3) = (1, 2)$, $T(1, -1) = (1, -1)$, and $T(4, 1) = (1, 0)$.

☐ If there is a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(2, 3) = (1, 2)$, and $T(1, -1) = (1, -1)$, then $T(x, y) = \frac{1}{5}(4x - y, -x + 4y)$.

Let $\beta = \{v_1, v_2, v_3\}$ and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be bases of a vector space V . Consider a linear transformation $T : V \rightarrow V$ such that $T(v_1) = 2v_3 + v_1$, $T(v_2) = 2v_1$, and $T(v_3) = 2v_2$. The matrix representation of T , with respect to β and γ for the domain and codomain respectively, is

☐
$$\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$$

Let $\{v_1, v_2, v_3\}$ be a basis of a vector space V and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be a basis of vector space W . Consider a linear transformation $T : V \rightarrow W$ such that $T(v_1) = 2v_3 + v_1$, $T(v_2) = 2v_1$, and $T(v_3) = 2v_2$. Then, the rank of T is 3.

☐

No, the answer is incorrect.

Score: 0

Accepted Answers:

If there is a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(2, 3) = (1, 2)$, and $T(1, -1) = (1, -1)$, then $T(x, y) = \frac{1}{5}(4x - y, -x + 4y)$.

Let $\beta = \{v_1, v_2, v_3\}$ and $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be bases of a vector space V . Consider a linear transformation $T : V \rightarrow V$ such that $T(v_1) = 2v_3 + v_1$, $T(v_2) = 2v_1$, and $T(v_3) = 2v_2$. The matrix representation of T , with respect to β and γ for the

domain and codomain respectively, is
$$\begin{bmatrix} 1 & \frac{2}{3} & \frac{2}{3} \\ 0 & 0 & 2 \\ 1 & -\frac{2}{3} & \frac{4}{3} \end{bmatrix}$$

Let $\{v_1, v_2, v_3\}$ be a basis of a vector space V and $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$ be a basis of vector space W . Consider a linear transformation $T : V \rightarrow W$ such that $T(v_1) = 2v_3 + v_1$, $T(v_2) = 2v_1$, and $T(v_3) = 2v_2$. Then, the rank of T is 3.

2) Consider the following set $S = \{(x, y) \mid x + y = 1, x, y \in \mathbb{R}\}$. In column A the matrix representation of some linear transformations are given with respect to the standard ordered bases. Match the entries in column A with the set $T(S)$ given in column B and the

1 point

geometric representations of S and $T(S)$ in column C.

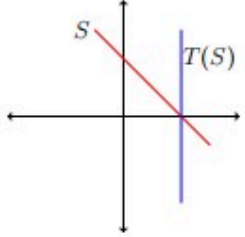
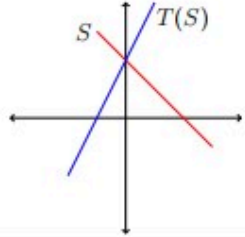
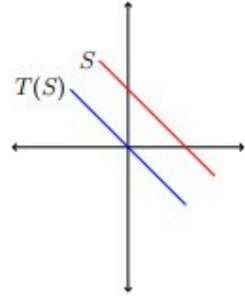
	Matrix form of linear transformation (Column A)		Image of the given set (Column B)		Geometric representations (Column C)
i)	$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$	a)	$T(S) = \{(x, y) \mid x = 1, y \in \mathbb{R}\}$	1)	
ii)	$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$	b)	$T(S) = \{(x, y) \mid x + y = 0, x, y \in \mathbb{R}\}$	2)	
iii)	$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$	c)	$T(S) = \{(x, y) \mid 2x - y = -1, x, y \in \mathbb{R}\}$	3)	

Table: M2W6P1

- ☐ i $\rightarrow$ b $\rightarrow$ 2.
- ☐ ii $\rightarrow$ c $\rightarrow$ 1.
- ☐ ii $\rightarrow$ c $\rightarrow$ 2.
- ☐ iii $\rightarrow$ a $\rightarrow$ 3.

☐ $i \rightarrow b \rightarrow 3.$

☐ $iii \rightarrow a \rightarrow 1.$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$ii \rightarrow c \rightarrow 2.$

$i \rightarrow b \rightarrow 3.$

$iii \rightarrow a \rightarrow 1.$

Multiple Select Questions (MSQ):

3) Consider the following two groups. In Group 1, there are 3 functions from one vector **1 point** space to another, and in Group 2, there are three properties of functions. All the vector spaces have usual addition and scalar multiplication.

Group-1

$$\mathbf{P}: T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ such that } T(x, y) = \begin{cases} (\frac{x^2}{y}, y) & \text{if } y \neq 0 \\ (x, 0) & \text{otherwise} \end{cases}$$

$$\mathbf{Q}: T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ such that } T(x, y) = (x, xy)$$

$$\mathbf{R}: T : \mathbb{R}^3 \rightarrow \mathbb{R} \text{ such that } T(x, y, z) = \frac{x+y+z}{5}$$

Group-2

$$1: T(v_1 + v_2) = T(v_1) + T(v_2) \text{ for all } v_1, v_2 \in V$$

$$2: T(cv) = cT(v) \text{ for all } v \in V \text{ and } c \in \mathbb{R}.$$

3: T is a linear transformation

Choose the set of correct options.

- ☐ P satisfies 1 but not 2.
- ☐ P satisfies 2 but not 1
- ☐ Q satisfies 1
- ☐ Q satisfies 3
- ☐ P satisfies 3
- ☐ R satisfies 2
- ☐ R satisfies 3
- ☐ R satisfies 1 but not 2

No, the answer is incorrect.

Score: 0

Accepted Answers:

P satisfies 2 but not 1

R satisfies 2

R satisfies 3

Numerical Answer Type (NAT):

4) Suppose the matrix representation of a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with respect to the ordered bases $\beta = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ for the domain and $\gamma = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ for the range, is $I_{3 \times 3}$, i.e., the identity matrix of order 3. Let A be the matrix representation of the linear transformation T with respect to standard ordered basis of $\mathbb{R}^3$ for both domain and range. Then the determinant of A is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

5) Let T and S be two linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$, which are defined as follows:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (x - y, y - z, z - x) \quad S(x, y, z) = (x + y, y + z, z + x)$$

Let $S + T$ be defined to be the linear transformation as follows:

$$(S + T) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$(S + T)(x, y, z) = S(x, y, z) + T(x, y, z)$$

Let C be the matrix representation of $S + T$ with respect to the standard ordered bases of $\mathbb{R}^3$. If $C = nI$, where I denotes the identity matrix of order 3, then what will be the value of n ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

6) Consider a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as $T(x, y, z) = (x + 5y, z - 3x, y + 6z)$. Which of the following options are true about T ? **1 point**

- ☐ Nullity of T is 0.
- ☐ A basis for the range of T is $\{(1, 5, 0), (0, 1, 6)\}$.
- ☐ A basis for the range of T is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.
- ☐ Rank of T is 2.

The matrix representation of T with respect to the standard ordered bases of

☐ $\mathbb{R}^3$ is $\begin{bmatrix} 1 & -3 & 0 \\ 5 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.

The matrix representation of T with respect to the standard ordered bases of

☐ $\mathbb{R}^3$ is $\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.

- ☐ Nullity of T is 1.
- ☐ Rank of T is 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Nullity of T is 0.

A basis for the range of T is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

The matrix representation of T with respect to the standard ordered bases of $\mathbb{R}^3$ is

$\begin{bmatrix} 1 & 5 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$.

Rank of T is 3.

7) Let V be a vector space with ordered basis $\alpha = \{v_1, v_2, \dots, v_{10}\}$. Consider a linear transformation $T : V \rightarrow V$ such that $T(v_1) = v_1$ and $T(v_i) = v_i - v_{i-1}$, where $i > 1$. Let A be the matrix representation of the linear transformation T , with respect to α for both the domain and codomain. what is the trace of A ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 10

1 point

Comprehension Type Question:

For each video on YouTube, consider the vector (Number of comments in thousands, Number of shares in thousands, Number of views in thousands) and consider the subset S of $\mathbb{R}^3$ consisting of these vectors. It is known that the ranking of a YouTube video can be thought of as the restriction of a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ to S where $\mathbb{R}^3$ comes with usual addition and scalar multiplication. (assume ranking 0 (zero) is possible and in practice negative rankings can be ignored).

The following data in Table M2W6P2 shows the correlation between comments, shares, and views - with YouTube rankings of a video.

Number of comments (in thousands)	Number of shares (in thousands)	Number of views (in thousands)	Ranking of a video
3	10	8	3
4	15	10	2
5	20	15	1

Table: M2W6P2

Use the above information to answer questions 8, 9 and 10.

8) Which of the following sets of vectors in S have the property that the YouTube video **1 point** corresponding to every linear combination of that set which is in S has ranking 0?

- ☐ $\{(1, 1, 0), (0, 0, 1)\}$
- ☐ $\{(6, 25, 0), (0, 0, 25)\}$
- ☐ $\{(25, 6, 0), (0, 0, 25)\}$
- ☐ $\{(12, 50, 0), (0, 0, 1)\}$
- ☐ $\{(24, 100, 0), (0, 0, 25)\}$
- ☐ $\{(6, 0, 0), (0, 25, 0), (0, 0, 1)\}$
- ☐ $\{(25, 0, 0), (0, 6, 0), (0, 0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(6, 25, 0), (0, 0, 25)\}$
 $\{(12, 50, 0), (0, 0, 1)\}$
 $\{(24, 100, 0), (0, 0, 25)\}$

9) Which of the following sets of vectors in S have the property that the YouTube video **1 point** corresponding to every linear combination of that set which is in S has ranking a multiple of 15?

- ☐ $\{(3, 0, 0), (0, 0, 1)\}$
- ☐ $\{(3, 0, 1)\}$
- ☐ $\{(3, 0, 1), (0, 0, 1)\}$
- ☐ $\{(0, 0, 1)\}$
- ☐ $\{(9, 25, 0), (0, 0, 1)\}$
- ☐ $\{(25, 9, 0), (0, 0, 1)\}$
- ☐

$\{(9, 0, 0), (0, 25, 0), (0, 0, 1)\}$

☐ $\{(25, 0, 0), (0, 9, 0), (0, 0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(3, 0, 0), (0, 0, 1)\}$

$\{(3, 0, 1)\}$

$\{(3, 0, 1), (0, 0, 1)\}$

$\{(0, 0, 1)\}$

$\{(9, 25, 0), (0, 0, 1)\}$

$\{(9, 0, 0), (0, 25, 0), (0, 0, 1)\}$

10) Suppose that for a YouTube video there are 5 (in thousands) shares. What must be the number of comments (in thousands) so that the ranking of the video will be 4? [Note: Suppose your answer is 6000, then you have to enter 6 as the answer.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Your score is: 0/10